MASTER DATA ENRICHMENT REPORT

Georgia EV Supply Chain Intelligence

Domain-Specific RAG Knowledge Base — Complete Step-by-Step Guide

What this report is: A complete, consolidated guide synthesized from 6 separate AI research responses (Claude, Perplexity, Gemini, ChatGPT Plus, ChatGPT Free, and a second Claude analysis). Every data source, every website, every search method, and every step to add data to your RAG pipeline is included here in one place.

Who it is for: You — a graduate researcher building a domain-specific RAG system for Georgia's EV supply chain, heading toward industry deployment.

1. Current Data Situation — The Problem You Are Solving

Before adding anything, it is important to be very clear about what your current data can and cannot answer. Every AI source reviewed agreed on the same core diagnosis:

Current Capability	Current Gap	Impact on RAG
Who exists (company names)	No investment amounts or dates	Cannot answer 'how much' or 'when'
Where they are located	26 companies missing OEM relationships	Cannot answer 'who supplies whom'
General product category	46-character avg. description	Retrieval fails on specific queries
Employment headcount	Only 2 battery cell manufacturers	Narrow battery coverage
Tier classification	No narrative or contextual text	No grounding for complex questions
EV relevance flag	No facility capacity data	Cannot answer production volume questions

The Core Goal:
Transform your system from answering 'Who exists?' to answering 'Who supplies whom, where, with what, how much, when, and how fast growing?' All 6 AI sources agreed this requires Events + Relationships + Narrative + Time.

2. Recommended Data Architecture — Build 5 Object Types

All sources agreed: do not try to make your current Excel file fatter. Instead build a multi-layer knowledge base with 5 distinct object types. This is the foundation everything else builds on.

Object Type	What It Contains	Example	Size per Chunk
Company Master Card	Canonical name, aliases, parent, county, EV role, OEM ties	SK On | Commerce GA | Battery Cell | Ford, VW	150–250 tokens

Event Record	Investment, expansion, permit, layoff, award, milestone	SK On $2.6B investment announced 2021	250–500 tokens
Facility/Permit Dossier	Process lines, capacity, regulatory info, site facts	SK On air permit: 21.5 GWh capacity, coating lines	300–700 tokens
Relationship Edge	Supplier→OEM edges with evidence text	F&P Georgia SUPPLIES SK On [Battery recycling]	1 sentence surrogate
Evidence Chunk	Narrative text from filings, press releases, reports	Full press release paragraph from georgia.org	400–700 tokens

Every object must carry these metadata fields: company_id, facility_id, county, source_url, source_type, publish_date, confidence, ttl_days (how long before it expires)

3. Complete Data Sources — Where to Go, What to Find, How to Get It

This section covers every source identified across all 6 AI research responses, organized by category with complete steps.

3.1 Georgia Government & Economic Development Sources

Priority: HIGH | Cost: Free | Best for: Investment amounts, dates, job counts, facility details

A. Georgia Department of Economic Development (GDEcD)

- **Main URL:** https://www.georgia.org/competitive-advantages/key-industries/electric-mobility
- **Press Releases:** https://www.georgia.org/press-release/ and https://gov.georgia.gov/press-releases
- **EV Ecosystem Page:** https://www.selectgeorgia.com/discover-georgia/industries/EV-ecosystem-in-georgia/
- **EMIA Report (Free PDF):** https://georgia.org/sites/default/files/2023-01/emia_report_-_22mcie121_12-2022.pdf
- **Hyundai Economic Development Agreement (Free PDF):** https://georgia.org/sites/default/files/2022-07/hyundai_motor_group_economic_development_agreement_-fully_signed_and_acknowledged.pdf
- **Hyundai Incentives Executive Summary:** https://georgia.org/sites/default/files/2023-09/hyundai_motor_group_incentives_-_updated_executive_summary_09.08.23_final.pdf

What to find: Every announced EV investment in Georgia — company name, investment dollar amount, job creation numbers, county, year, facility type, OEM customer, production capacity

Step-by-step process:

1. Go to georgia.org/press-release and search: electric vehicle, battery, EV plant, Hyundai, Kia, Rivian, SK
2. For each press release found, copy the full text into a new row in your KB as Extended_Description
3. Extract structured fields: company, investment_amount, jobs, county, date, facility_type, oem_customer
4. Download the EMIA report PDF and extract the company list from Figure 5 — it contains EV role labels
5. Download the Hyundai economic development agreement — it contains exact investment obligations and affiliate lists

Gap filled: Investment amounts (your biggest gap), dates, job counts, OEM relationships, and narrative facility descriptions

B. Savannah Harbor–Interstate 16 Corridor Joint Development Authority (JDA)

- **HMGMA Suppliers List:** https://www.savannahjda.com/announced-hmgma-suppliers/
- **Bryan County Project Page:** https://www.bryancountyga.org/economic-development

What to find: Every announced supplier for the Hyundai Metaplant — company name, date announced, jobs, component supplied. This is a ready-made supplier relationship list.

Step-by-step process:

6. Go to savannahjda.com/announced-hmgma-suppliers and scrape or manually copy the full table
7. For each supplier listed, create a relationship edge: [Supplier] SUPPLIES [HMGMA] with component and date
8. Cross-reference each supplier name against your GNEM file to update missing OEM relationship data
9. For suppliers NOT in your GNEM file, add them as new rows

Why this is critical: This single page solves your missing OEM relationship problem for the entire Hyundai ecosystem. It contains $2.5+ billion in supplier announcements with exact component details (EV Power Electric systems, Integrated Charging Control Units, thermal management, etc.)

C. County-Level Development Authorities

Target the counties with the highest EV activity:

County	Key EV Company	JDA / GIS URL
Bryan County	Hyundai HMGMA	bryancountyga.org/economic-development + bryancountyga.gov/government/departments-a-g/gis-mapping
Troup County	Kia Georgia, Hyundai Transys	troupe county development authority website
Jackson County	SK On (SK Battery America)	jacksoncountygov.com/256/GIS-Addressing
Bartow County	Hyundai-SK On JV Battery	cartersville-bartow.com economic development
Morgan/ Walton/Newton	Rivian	East Atlanta Megasite / Stanton Springs North JDA

Step-by-step: For each county, find their economic development page and scrape meeting minutes PDFs, project pages, and incentive announcements. Use pdfplumber in Python to extract tables from PDFs.

D. Georgia EPD — Environmental Permits (Reveals Facility Capacity)

- **Air Permit Search:** https://permitsearch.gaepd.org/
- **GEOS Public Portal:** https://geos.epd.georgia.gov/ga/geos/public/
- **PSD/NSR Permit Database:** https://epd.georgia.gov/psd112gnaa-nsrpcp-permits-database
- **EPD Forms & Permits:** https://epd.georgia.gov/forms-permits

Why permits are valuable: Environmental permits legally require companies to disclose their 'Maximum Hourly Design Rate' for every piece of equipment. A battery plant's air permit will explicitly list cathode coating line capacity, solvent recovery units, GWh output, and operational status. This is often the only public source for actual production capacity.

Step-by-step:

10. Go to permitsearch.gaepd.org
11. Search each of your 42 EV-relevant companies by name and county
12. Download any Title V or PSD permit PDFs found

13. Use Python pdfplumber to extract the 'Equipment List' and 'Process Description' tables
14. Convert extracted tables to Markdown format (Markdown tables retain structure better in embeddings)
15. Add as Facility Dossier objects with metadata: permit_number, issue_date, facility_id, county

E. Georgia Secretary of State — Business Registry

- **Business Search:** https://ecorp.sos.ga.gov/BusinessSearch

What to find: Legal entity names, registration status, control numbers, registered agents, formation date, officer names, parent/subsidiary hints

Step-by-step:

16. For every company in your GNEM file, run a name search on GA eCorp
17. Capture: legal_name, control_number, status (active/inactive), principal_office_address, registered_agent
18. Compare legal name to GNEM name — many companies have different legal vs. trade names
19. Use control numbers to match across other databases and catch duplicates
20. Add legal_name, aliases, and control_number to your company master card

F. Georgia Environmental Finance Authority (GEFA)

- **Main URL:** https://gefa.georgia.gov
- **GRIP Grant (Free):** https://gefa.georgia.gov — search Grid Resilience

What to find: EV charging infrastructure grants, clean energy projects, grid improvement investments totaling $507M+ that support EV manufacturing counties

3.2 Federal Government Sources

> **Priority: HIGH | Cost: Free | Best for: Federal funding, facility-level regulatory data, trade statistics**

A. SEC EDGAR — Annual Reports of Public Companies

- **Search:** https://www.sec.gov/cgi-bin/browse-edgar
- **API:** https://data.sec.gov/

Companies to search: Kia (parent: Hyundai Motor Co), SK On (parent: SK Innovation), Hanwha Q Cells, ZF Friedrichshafen, Volvo, Yamaha Motor, Hitachi Astemo

Step-by-step:

21. Go to sec.gov/cgi-bin/browse-edgar and search the parent company name
22. Select the most recent 10-K (annual) or 20-F (foreign annual) filing
23. Press Ctrl+F inside the filing and search for: Georgia, Commerce, Bryan County, West Point, battery, electric vehicle
24. Copy every paragraph mentioning Georgia operations — these contain investment amounts, facility sizes, capacity, and OEM customer names
25. Store each extracted paragraph as an evidence chunk with metadata: company, form_type, fiscal_year, section

B. US Department of Energy — Grants and Loans

- **Battery Manufacturing Grants:** https://www.energy.gov/cmei/manufacturing/battery-manufacturing-and-recycling-grants
- **Battery Materials Processing Grants:** https://www.energy.gov/cmei/manufacturing/battery-materials-processing-grants

- **Loan Programs Office Portfolio:** https://www.energy.gov/lpo/portfolio-projects
- **DOE NEPA Search (Environmental Assessments):** https://www.energy.gov/nepa/articles

Step-by-step:

26. On each DOE page, use Ctrl+F to search for Georgia
27. For each Georgia project found, copy the project description, grant amount, company, county, technology type
28. Check NEPA pages for environmental assessments of Georgia battery projects — these contain facility details
29. Notable Georgia recipients to search: Ascend Elements, Anovion, Solvay, Syrah Resources

C. EPA ECHO — Facility Cross-Reference

- **Facility Search:** https://echo.epa.gov/facilities/facility-search
- **RCRAInfo (Hazardous Waste):** https://rcrapublic.epa.gov/rcra-hwip/search

Step-by-step:

30. Search each GNEM company by name and address in EPA ECHO
31. Match found facilities to your GNEM records using FRS identifier
32. Collect program IDs, compliance status, and hazardous-waste handler status
33. Companies with hazardous waste handling are confirmed manufacturing operations, not offices — use this to validate your EV relevance flags

D. USAspending — Federal Contract and Grant Awards

- **Recipient Search:** https://www.usaspending.gov/recipient
- **API:** https://api.usaspending.gov/
- **SAM Contract Awards:** https://sam.gov/fpds

Step-by-step:

34. Search each company name on usaspending.gov
35. Filter by Georgia and EV/battery/charger/thermal keywords
36. Capture: agency name, award amount, NAICS code, description, period of performance
37. This reveals Georgia companies working on EV buses, charging hardware, grid equipment for government customers

E. Alternative Fuels Data Center (AFDC)

- **Main URL:** https://afdc.energy.gov
- **Free API:** https://developer.nrel.gov/docs/transportation/alt-fuel-stations-v1/
- **Georgia NEVI Deployment Map:** https://experience.arcgis.com — search Georgia NEVI Deployment Map 2024

Step-by-step:

38. Use the API with parameters: state=GA, fuel_type=ELEC
39. Download the Georgia NEVI data as GeoJSON from the ArcGIS hub
40. Use company locations to compute which EV facilities are near NEVI corridors
41. Add corridor proximity as metadata to facility records

F. US Bureau of Labor Statistics — Employment Data

- **QCEW Open Data:** https://www.bls.gov/cew/additional-resources/open-data/
- **Georgia Labor Market Explorer:** https://explorer.gdol.ga.gov/

Step-by-step:

42. Go to BLS QCEW and filter by state=Georgia, NAICS codes: 3363 (Motor Vehicle Parts), 3359 (Electrical Equipment), 3353 (Electrical Equip Mfg)
43. Download county-level employment counts by NAICS
44. Use this data to answer questions like 'how many automotive workers are in Jackson County?'
45. Store as a regional_stats reference table, not narrative chunks

3.3 Supplier Relationship and Network Sources

Priority: CRITICAL | Cost: Mix of free and paid | Best for: Who supplies whom, OEM relationships, component-level connections

A. MarkLines — Automotive Supplier-OEM Database

- **Supplier Database:** https://www.marklines.com/en/supplier_db
- **Who Supplies Whom:** https://www.marklines.com/en/wsw/search

What to find: Exact supplier-to-OEM relationships with part families (battery pack, thermal management, BMS, wiring, stamping, electronics). This is the single best source for answering 'which suppliers serve Kia Georgia?'

Step-by-step:

46. Search by OEM: Kia, Hyundai, Genesis, SK On
47. Filter by part family relevant to EVs
48. Export relationship triples: Supplier → Part → OEM/Model
49. Store as graph edges with one-sentence text surrogate for BM25 retrieval
50. Note: some content requires free registration; deeper data may require subscription

B. ImportYeti — Free Shipment Intelligence

- **Main URL:** https://www.importyeti.com/

What to find: Bill of lading records showing which Georgia companies are importing from overseas suppliers. This reveals actual supply chain relationships that press releases often omit.

Paid alternatives for deeper data: Panjiva (spglobal.com/market-intelligence), PIERS, CargoFax (cargofax.co), Volza (volza.com), Zauba (zauba.com)

Step-by-step:

51. Go to importyeti.com and search each company name from your GNEM file
52. Look for shipment records with HS codes: 8507.60 (lithium-ion batteries), 8507.90 (battery parts), 2836.91 (lithium carbonate), 2827.39 (battery electrolyte materials)
53. Extract: shipper, consignee, product description, port, date
54. Use your local LLM (Qwen) to synthesize raw shipment data into relationship narratives
55. Store as relationship edges and trade_profile facts, not raw shipment rows

C. OEM Official Supplier Pages

OEM	Supplier Information URL	What You Find
Hyundai Motor Group	hyundai.com/worldwide/en/newsroom	Supplier announcements, affiliate lists
Kia Georgia	kiageorgia.com/category/news/	Plant news, supplier relationships
SK On	skon.co + SEC EDGAR SK Innovation	Capacity, customers, JV details
HMGMA	hmgma.com	All Metaplant details and affiliate companies

Rivian	rivian.com + SEC EDGAR (public)	Georgia campus, suppliers, investment

Step-by-step:

56. For each OEM, find their newsroom or press releases page
57. Search for 'supplier', 'Georgia', 'battery', 'component'
58. Extract all mentioned supplier names and what they supply
59. Cross-reference against your GNEM file to update missing OEM relationship data

3.4 Industry Databases and Registries

Priority: HIGH | Cost: Free (some paid) | Best for: Company discovery, capability details, sector coverage

A. Georgia Supply Chain Exchange (GaMEP) — FREE DATABASE

- **URL:** https://gscx.gamep.org

What to find: Free database of Georgia manufacturers with capability listings, NAICS codes, processes, certifications, and primary industries. Far richer than your current 46-character descriptions.

Step-by-step:

60. Create a free account on gscx.gamep.org
61. Search for EV-relevant keywords: battery, power electronics, thermal management, stamping, lightweight materials, HV harness, wiring
62. Export or copy capability profiles for each matched company
63. Use LLM to standardize extracted text into structured fields: processes, materials, certifications, EV_relevance
64. Merge into your GNEM KB via fuzzy name and location matching

B. GNEM E-Mobility Directory

- **URL:** https://emobility.uga.edu/

What to find: Members of the Georgia Network for Electric Mobility including utilities, software firms, research institutions, and manufacturers not in your GNEM Excel file. Also contains the membership network map showing organization relationships.

Step-by-step:

65. Go to emobility.uga.edu and find the E-Mobility Directory and Membership Network Map
66. Scrape all listed organization names and types
67. Compare against your 205-company GNEM list — add any new companies as new rows
68. Add network_membership field to existing companies: GNEM, GAMA, MAGNET

C. Georgia Automotive Manufacturers Association (GAMA)

- **URL:** Search 'GAMA Georgia Automotive Manufacturers Association' — GNEM announcement links to it

What to find: Member directory of OEMs, Tier 1/2 suppliers in Georgia including companies serving Kia Georgia, Blue Bird, Hyundai

D. US and Canada EV Supply Chain Map (Big Green Machine)

- **URL:** Charge search 'US Canada Electric Vehicle Supply Chain Map Big Green Machine'

What to find: National EV and battery investment database with filters for Georgia — includes investment amounts, GWh capacity, battery segment (cathode, anode, cell, pack, recycling), and government support type

Step-by-step:

69. Find the Big Green Machine dashboard and request access to underlying dataset
70. Filter rows where state == GA
71. Extract: company, project, technology, investment, capacity (GWh), announced date, status, policy support
72. Add technology_segment field: cathode active material, pack assembly, e-axle, etc.

E. Thomasnet — Manufacturer Discovery

- **Georgia Manufacturers:** https://www.thomasnet.com/suppliers/georgia/

What to find: Tier 2/3 manufacturers in Georgia that your GNEM file may have missed — especially smaller shops doing metal stamping, plastics, wiring harnesses, and electronics

Step-by-step:

73. Search Thomasnet with categories: coolant hose, battery tray, stamped body part, wiring harness, EV connector, charger enclosure — filter by Georgia location
74. Label found companies as candidate_supplier records with confidence score
75. Validate automotive relevance by checking their company website before adding to KB

F. MEMA — Motor Equipment Manufacturers Association

- **URL:** https://www.mema.org/membership/original-equipment-suppliers

What to find: Member directory of OEM suppliers — use to validate tier classifications and find Georgia members not in your dataset

3.5 Company-Level Direct Research

> **Priority: HIGHEST | Cost: Free | Best for: Fixing the 46-character description problem immediately**

For each of your 42 directly EV-relevant companies, follow this exact process to collect a rich Extended_Description:

> **Company Research Process — Do This For Each of Your 42 EV Companies:**
> Step 1: Google '[Company Name] Georgia electric vehicle facility'
> Step 2: Go to their official website → find About Us, Facilities, Newsroom, or Press Releases
> Step 3: Search for 'Georgia' on their site
> Step 4: Copy the most relevant 2-3 paragraphs about their Georgia operations
> Step 5: Also search georgia.org/press-release for the GDEcD announcement about this company
> Step 6: Add the combined text as 'Extended_Description' column in kb_master.xlsx
> Step 7: Also extract: investment_amount, jobs, year_announced, production_capacity, oem_customers

Company	Georgia Location	Official Website	GDEcD Search Term
Kia Georgia Inc.	West Point, Troup County	kiageorgia.com	Kia Georgia West Point
Hyundai HMGMA	Ellabell, Bryan County	hmgma.com	Hyundai Metaplant Georgia
SK On (SK Innovation)	Commerce, Jackson County	skon.co	SK Battery America Commerce

Hitachi Astemo Americas	Shorter, Coweta County	hitachiastemo.com	Hitachi Astemo Georgia
Honda D&M	Tallapoosa, Cherokee County	hondainamerica.com	Honda Development Georgia
F&P Georgia Mfg	Rome, Floyd County	fp-group.com	F&P Georgia battery
Hollingsworth & Vose	Hawkinsville, Bleckley County	hovo.com	Hollingsworth Vose Georgia
Hyundai Transys Georgia	West Point, Troup County	hyundai-transys.com	Hyundai Transys West Point
ZF Gainesville LLC	Gainesville, Hall County	zf.com/us	ZF Gainesville Georgia
Yamaha Motor Mfg	Newnan, Coweta County	global.yamaha-motor.com	Yamaha Motor Newnan Georgia

3.6 Trade, Logistics, and Port Data

Priority: MEDIUM | Cost: Mix of free and paid | Best for: Supply chain flows, logistics context, new supplier discovery

A. Georgia Ports Authority

- **Main URL:** https://www.gaports.com
- **RO/RO Cargo Data:** https://gaports.com/ro-ro/ — Colonel's Island Terminal in Brunswick handles finished EVs

What to find: Vehicle throughput (Record 86,577 RO/RO units in May 2024), capacity expansion ($262M expansion to 120 acres, 1.5M annual capacity), inbound battery materials, logistics partners

B. USA Trade Online — Census Trade Data

- **URL:** https://usatradeonline.census.gov/

Step-by-step:

76. Filter by state = Georgia, NAICS codes for EV-related manufacturing
77. Extract aggregate import/export values for battery components
78. Store as regional_stats context, not as company-level facts

3.7 Geospatial and Infrastructure Sources

Priority: MEDIUM | Cost: Free | Best for: Location intelligence, spatial queries, facility expansion signals

Source	URL	What to Get
Georgia GIS Clearinghouse	gisdata.georgia.gov	Industrial zones, county boundaries, roads
Bryan County GIS	bryancountyga.gov/government/departments-a-g/gis-mapping	Parcel polygons for HMGMA area
Jackson County Open Data	jackson-county-ga-open-data-portal-jacksoncountyga.hub.arcgis.com	Parcel data for SK On area
OpenStreetMap	overpass-turbo.eu	Industrial building footprints, rail spurs

Overpass API		
Georgia State Imagery	Georgia.gov GIS Imagery Hub	Satellite imagery for construction progress
GDOT NEVI Map	experience.arcgis.com — Georgia NEVI 2024	EV corridor locations and planned stations

How to use geospatial data in RAG: Do NOT embed raw coordinates into ChromaDB. Instead, derive text descriptions like 'Located 12 miles from HMGMA', 'On the I-16 EV corridor', 'Adjacent to rail spur' and add these as text fields in facility records.

3.8 Academic and Research Publications

Priority: MEDIUM | Cost: Free | Best for: Technical context, industry analysis, domain background

Source	URL	Search Query to Use
Google Scholar	scholar.google.com	"Georgia" "electric vehicle" "supply chain" "manufacturing"
UGA Research	terry.uga.edu/research + selig.uga.edu	Georgia automotive economic impact 2024
Georgia Tech ISyE	isye.gatech.edu	Georgia EV supply chain optimization
NREL EV Reports	nrel.gov/transportation/ev-market-reports.html	Georgia battery manufacturing
DOE Critical Materials	energy.gov	Critical materials battery supply chain Georgia

3.9 Real-Time Monitoring Sources

Priority: HIGH for Production | Cost: Free | Best for: Keeping KB current for industry users

Tool	URL	Queries to Set Up
Google Alerts	google.com/alerts	"Georgia" "electric vehicle" investment; "battery plant Georgia"; "EV supplier Georgia announcement"
Automotive News	autonews.com	Search: Georgia EV plant, Georgia battery, Kia West Point, SK On Commerce
Atlanta Business Chronicle	bizjournals.com/atlanta	EV, battery, automotive supplier, Georgia investment
The Current GA	thecurrentga.org	Georgia EV supply chain news — local investigative coverage
LinkedIn Company Pages	linkedin.com/company/[name]	Check each of your 42 companies' newsfeeds monthly

4. Data Enrichment Techniques — How to Make Thin Data Rich

These techniques transform existing sparse data and turn it into rich, retrievable knowledge. All 6 AI sources identified these as essential.

4.1 Extended Description Paragraphs (Do This First — Highest Impact)

What it is: Replacing 46-character product descriptions with 200-400 word narrative paragraphs about each company's Georgia operations.

Before (Current KB): Product / Service: Battery cells for EV

After (With Extended Description): SK On, formerly SK Innovation's battery division, operates a lithium-ion battery cell manufacturing facility in Commerce, Jackson County, Georgia. The plant, announced in 2021 with a $2.6 billion investment, produces battery cells primarily for Ford F-150 Lightning and Volkswagen electric vehicles. The facility employs approximately 2,600 workers and has an annual production capacity of 21.5 GWh. SK On selected Georgia due to its existing automotive supply chain ecosystem, proximity to OEM customers, and state incentive packages including tax credits from the Georgia Department of Economic Development. Production began in 2022 making it one of the first dedicated EV battery cell plants in the Southeast United States.

How to add to code: In src/kb_loader.py, add 'Extended_Description' to the KB_TEXT_FIELDS list. That is the only code change needed.

4.2 LLM-Based Synthetic Profile Generation

What it is: Using your local Qwen or Gemma model to automatically expand thin descriptions by searching and summarizing web content for each company.

Step-by-step:

79. For each company with a thin description, use your Tavily API to search: '[Company Name] Georgia electric vehicle facility operations'
80. Feed the Tavily results to your local Qwen model with this prompt:

Prompt template to use with Qwen: "You are a Georgia EV supply chain analyst. Based ONLY on the provided web search results, write a 300-word structured profile for [COMPANY NAME] describing: (1) their Georgia facility location and size, (2) their specific EV supply chain role, (3) their main products and technologies, (4) their OEM customers, and (5) investment/employment details. If information is not in the search results, say 'Not available from public sources'. Do not invent facts."

81. Store the generated profile as Extended_Description with a flag: generated=true, evidence_sources=[URLs]
82. Always keep the original sparse description unchanged alongside the generated version

4.3 Supplier Relationship Graph Construction

What it is: Extracting supplier-OEM relationships from text and storing them as graph edges — the missing structural layer your flat Excel KB cannot provide.

Step-by-step:

83. Collect all press releases, JDA announcements, and OEM newsroom content mentioning Georgia suppliers
84. Run each document through Qwen with this extraction prompt:

Prompt: "Extract all supplier-customer relationships from this text. For each relationship found, output a JSON object: {supplier: '[name]', supplies_to: '[OEM name]', component: '[what is supplied]', location: '[Georgia county if mentioned]', source_sentence: '[exact quote]', confidence: 'high/medium/low'}. Only extract relationships explicitly stated in the text."

85. Store extracted triples as relationship edges in a separate collection
86. Add a one-sentence text surrogate per edge so BM25 can find them
87. For production: build these into a Neo4j or NetworkX graph for multi-hop queries

4.4 Investment and Timeline Data Collection

Add a structured Investment_Details column to your KB with these fields for each company:

Field	Example Value	Where to Find It
announcement_year	2021	georgia.org press releases
investment_amount	$2.6 billion	GDEcD announcements, SEC filings
jobs_announced	2600	Governor's office press releases
production_start_year	2022	Company newsroom, GDEcD
annual_capacity	21.5 GWh	Air permits, DOE filings
incentive_type	Georgia tax credits + county incentives	Economic development agreements
oem_customers	Ford, Volkswagen	Press releases, SEC filings

4.5 PDF Extraction for Reports and Permits

What to install: pip install pdfplumber pymupdf camelot-py tabula-py

Step-by-step process for any PDF:

88. Try pdfplumber first for digital PDFs with selectable text
89. Use camelot for PDFs with visible table lines
90. Use pymupdf for complex layouts
91. For scanned PDFs (images), use Google Cloud Document AI or Docling for layout-aware extraction
92. Convert extracted tables to Markdown format before embedding — Markdown tables preserve relationships better
93. Chunk PDFs by named section, not by page: permit_header, process_units, capacity_table, compliance_dates

Priority PDFs to download and process:

- EMIA Report: georgia.org/sites/default/files/2023-01/emia_report_-_22mcie121_12-2022.pdf
- Hyundai Economic Development Agreement: (URL in Section 3.1 above)
- Georgia EPD air permits for SK On, Hyundai HMGMA, Honda, F&P Georgia
- DOE NEPA Environmental Assessments for Georgia battery projects
- Georgia Power Automotive Manufacturing Report: selectgeorgia-production-assets.s3.amazonaws.com

4.6 Patent Data for Technology Context

- **URL:** https://patents.google.com and https://patentsview.org

Step-by-step:

94. Search '[Company Name] battery Georgia' or '[Company Name] electric vehicle thermal'
95. Download patent abstracts (100-200 words each) for relevant hits
96. Store as Technology_Context field in your KB
97. Add CPC classification tags: B60L (EV powertrains), H01M (electrochemical cells), F28D (thermal management)

4.7 Job Posting Intelligence

- **Sources:** LinkedIn Jobs (linkedin.com/jobs), Indeed (indeed.com), company career pages

Step-by-step:

98. For each of your 42 EV companies, search their company name + Georgia on LinkedIn Jobs and Indeed
99. Filter for job titles with: battery, electric vehicle, BMS, thermal, HV harness, cathode, anode
100. Extract implied products and technologies from job descriptions
101. Add a Recent_Activity field: '2024: Hiring Battery Systems Engineers and Electrode Process Technicians — indicates active battery cell manufacturing expansion'
102. Refresh this check quarterly

5. Complete Search Query Templates

Use these exact queries. Copy and paste them into Google, Tavily, or specific site search bars.

5.1 Google Search Queries

For Individual Companies (replace [COMPANY]):
[COMPANY] Georgia facility electric vehicle 2022 2023 2024
[COMPANY] Georgia plant investment announcement jobs
[COMPANY] battery manufacturing Georgia capacity GWh
[COMPANY] annual report Georgia operations electric vehicle
site:georgia.org [COMPANY] press release
site:sec.gov [PARENT COMPANY] 10-K Georgia battery

For Finding New Companies Not in GNEM:
Georgia electric vehicle supplier investment 2024 announced
Georgia battery manufacturing plant groundbreaking
"Jackson County" OR "Bryan County" OR "Troup County" electric vehicle
Georgia Tier 1 automotive supplier EV battery 2023 2024
site:georgia.org electric vehicle announcement 2024

For Supplier Relationships:
[COMPANY] supplier to Kia Georgia West Point
[COMPANY] supplier Hyundai HMGMA Bryan County
[COMPANY] OEM customer Georgia automotive
who supplies thermal management Hyundai Metaplant Georgia
HMGMA tier 1 suppliers Bryan County announced

5.2 Tavily Batch Query Templates for Your Web Crawler

Intent	Tavily Query	Gap Filled
Investment amounts	SK On Commerce Georgia battery investment billion 2021	Temporal + financial data
Plant capacity	Kia Georgia West Point annual production capacity EVs	Production volumes
New entrants	Georgia new electric vehicle supplier 2024 2025 announcement	New GNEM companies
HMGMA suppliers	Hyundai Metaplant Bryan County tier 1 suppliers announced	OEM relationships
Battery technology	Georgia battery cell manufacturing lithium ion technology GWh	Technical context
Employment updates	Georgia automotive supplier hiring expansion jobs 2024	Updated headcounts
Charging infrastructure	Georgia NEVI EV charging infrastructure deployment 2024	Charging network
Policy context	Georgia EV tax incentive manufacturing supplier 2024	Policy/regulatory
Recycling sector	Georgia battery recycling Ascend Elements circular economy	Missing sub-sector
Federal grants	DOE battery grant Georgia Anovion Solvay Syrah	Federal funding data

6. Exact Code Changes Needed in Your Project

All data enrichment requires only minimal code changes. Here is exactly what to change:

6.1 src/kb_loader.py — Add New Fields

```
Add these fields to KB_TEXT_FIELDS list:
KB_TEXT_FIELDS = [
    # Existing fields (keep all):
    'Company', 'Category', 'Industry Group', 'Updated Location',
    'Address', 'Primary Facility Type', 'EV Supply Chain Role',
    'Primary OEMs', 'Supplier or Affiliation Type', 'Employment',
    'Product / Service', 'EV / Battery Relevant', 'Classification Method',
     # NEW FIELDS — add these:
    'Extended_Description',    # 200-400 word narrative paragraph
    'Investment_Details',      # investment amount, year, jobs, capacity
    'Supplier_Context',        # OEM relationship description
    'Recent_Activity',         # current hiring / expansion signals
    'Technology_Context',      # from patents and technical sources
]
```

6.2 config/config.yaml — Add Domain Keywords

```
Add to web_validator.domain_keywords.tier1:
    - georgia economic development
    - jackson county georgia
    - bryan county georgia
    - troup county georgia
    - sk on commerce
    - kia georgia west point
    - hmgma hyundai metaplant
    - rivian normal georgia
    - battery gigafactory georgia
    - savannahjda
    - sk battery america
    - ascend elements georgia
```

6.3 New Python Script — PDF Text Extractor

```
Install:  pip install pdfplumber pymupdf camelot-py
# pdf_to_kb.py — extract text from PDFs and add to KB
import pdfplumber import pandas as pd  def extract_pdf_to_kb(pdf_path, company_name,
source_url, doc_type):
    chunks = []     with pdfplumber.open(pdf_path) as pdf:
        for page in pdf.pages:            text = page.extract_text()
            if text and len(text) > 100:                chunks.append({
                    'Company': company_name,                    'Extended_Description': text[:2000],
                    'EV / Battery Relevant': 'Yes',                    'Classification Method': 'PDF_Extract',
                    'Product / Service': f'{doc_type} — {source_url}'                })
    return chunks
```

7. Phased Implementation Plan

Phase 1 — Quick Wins (2–3 Days, No Advanced Coding Needed)

> **Expected impact: +15-25% answer correctness for EV-specific questions**

103. Add Extended_Description column to kb_master.xlsx — collect 2-3 paragraph descriptions for your 42 EV companies using georgia.org press releases and company websites
104. Fill 26 missing OEM relationships — use savannahjda.com supplier list for Hyundai ecosystem, OEM newsrooms for others
105. Add 5 new companies as new rows: Rivian (Normal, Morgan County), HMGMA, Ascend Elements, FREYR Battery, Kolon Industries
106. Update src/kb_loader.py — add Extended_Description to KB_TEXT_FIELDS (one line change)
107. Download and add the EMIA report PDF — extract the company list and role descriptions

Phase 2 — Structured Enrichment (1 Week)

> **Expected impact: +25-40% improvement in context recall and relationship questions**

108. Collect investment data from GDEcD press releases — add Investment_Details column for top 20 EV companies
109. Download and process 5 priority PDFs: EMIA report, Hyundai Economic Development Agreement, Georgia Power Automotive Report, NREL EV Report, UGA Economic Impact Study
110. Build supplier relationship edges — run LLM extraction over JDA announcements and OEM press releases
111. Research Georgia EPD permits for top 10 facilities — add capacity and process data
112. Set up Google Alerts for continuous monitoring: 'Georgia electric vehicle', 'battery plant Georgia', 'EV supplier Georgia'

Phase 3 — Deep Intelligence (2–4 Weeks)

> **Expected impact: Full production-ready KB with relationship reasoning capability**

113. Build LLM synthetic profile generator — expand all 42 EV companies using Tavily + Qwen pipeline
114. Add patent data for technology context — search Google Patents for your 42 EV companies
115. Integrate Georgia Secretary of State registry — canonical names, aliases, legal entity resolution
116. Add county-level employment data from BLS QCEW — enables aggregate county questions
117. Implement scheduled background crawler — weekly Tavily queries using templates from Section 5.2
118. Add job posting intelligence — quarterly review of LinkedIn/Indeed for each company

Phase 4 — Production Architecture (1–2 Months)

119. Implement graph layer — store supplier relationships in NetworkX or Neo4j alongside ChromaDB
120. Add 4-collection architecture: kb_company_master, kb_events, kb_relationships, kb_evidence_chunks
121. Implement freshness scoring — bias retrieval toward recent records for investment and capacity questions
122. Add continuous monitoring pipeline with deduplication via embeddings
123. Implement hybrid query routing — structured graph queries for relationship questions, vector search for narrative questions

8. Master Source Reference Table

All sources ranked by priority with exact URLs, type, cost, and what gap each fills:

#	Source	URL	Priority	Cost	Gap Filled
1	GDEcD Press Releases	georgia.org/press-release	Highest	Free	Investments, dates, jobs
2	HMGMA Suppliers (JDA)	savannahjda.com/announced-hmgma-suppliers	Highest	Free	OEM relationships (Hyundai)
3	SEC EDGAR	sec.gov/cgi-bin/browse-edgar	Highest	Free	Capacity, financials, SEC data

4	Georgia EPD Permits	permitsearch.gaepd.org	Highest	Free	Facility capacity, processes
5	Company Websites	Each company.com	Highest	Free	Fix 46-char descriptions
6	DOE Battery Grants	energy.gov/cmei/ manufacturing	Highest	Free	Federal funding, R&D
7	EMIA Report (PDF)	georgia.org (see URL in 3.1)	Highest	Free	Company roles, EV segments
8	GA Secretary of State	ecorp.sos.ga.gov	High	Free	Legal names, subsidiaries
9	EPA ECHO	echo.epa.gov	High	Free	Facility validation
10	USAspending	usaspending.gov	High	Free	Government contracts
11	GaMEP Supply Chain Exchange	gscx.gamep.org	High	Free	Capability descriptions
12	GNEM E-Mobility Directory	emobility.uga.edu	High	Free	New companies, network
13	Google Patents	patents.google.com	High	Free	Technology context
14	Automotive News	autonews.com	High	Free (news)	Investment news, plants
15	ImportYeti	importyeti.com	High	Free	Supply chain flows
16	Big Green Machine	US EV Supply Chain Map	High	Free request	Investment database
17	LinkedIn Jobs	linkedin.com/jobs	High	Free	Operational signals
18	Google Alerts	google.com/alerts	High	Free	Real-time updates
19	Georgia Ports Authority	gaports.com	Medium	Free	Logistics, export data
20	BLS QCEW	bls.gov/cew	Medium	Free	County employment
21	AFDC / NEVI ArcGIS	afdc.energy.gov + ArcGIS	Medium	Free	EV infrastructure
22	Thomasnet	thomasnet.com/suppliers/ georgia	Medium	Free	Tier 2/3 discovery
23	Google Scholar	scholar.google.com	Medium	Free	Academic context
24	MarkLines	marklines.com/en/ supplier_db	High	Paid/ Free tier	OEM-supplier relationships
25	Panjiva / PIERS	spglobal.com/market-intelligence	Medium	Paid	Deep trade intelligence

9. Advanced and Unique Data Ideas from AI Research

These ideas were mentioned across multiple AI sources and represent approaches beyond basic web search:

9.1 Georgia Open Records Act Requests

Any citizen can request government documents under Georgia's Open Records Act. For your project this means you can formally request the actual incentive agreements, tax abatement terms, and project details that development authorities have on file for EV companies. These documents contain exact investment obligations, clawback clauses, milestone dates, and job targets not available in press releases.

- Contact: County development authorities for Bryan, Jackson, Troup, Bartow Counties
- Request: 'All project agreements and incentive documents related to [Company Name] facility'
- Timeline: Georgia law requires a response within 3 business days

9.2 Utility Infrastructure as a Manufacturing Signal

- **Georgia Power EV Report:** selectgeorgia-production-assets.s3.amazonaws.com — search 'Automotive Manufacturing Georgia Power 2024'
- **GEFA Grid Intelligence:** gefa.georgia.gov — $507M GRIP grid upgrade for 38 Georgia EMCs serving EV manufacturers

A company requesting a 50MW power connection to Georgia Power is a confirmed large-scale manufacturer. Power interconnection requests are public record through the Georgia Public Service Commission and reveal factory scale before construction is complete.

9.3 The Ray Highway and Connected Vehicle Data

- **URL:** theray.org and cvprogram-gdot.hub.arcgis.com (GDOT Connected Vehicle Program)

The Ray is an 18-mile EV testing corridor on I-85 in West Georgia. Kia Georgia equipped 10 vehicles with on-board units for this corridor. This is a unique Georgia-specific EV intelligence source that reveals which manufacturers are testing connected vehicle technology.

9.4 Savannah JDA Supplier Matrix — The Most Underused Source

The Savannah Harbor-Interstate 16 JDA maintains a continuously updated list of every announced HMGMA supplier with date, company, jobs, and component supplied. This single page directly solves your OEM relationship gap for the entire Hyundai ecosystem and represents $2.5+ billion in announced supplier investments. This was identified as a high-priority source by both Claude responses and the Gemini response.

9.5 Synthetic Q&A Generation for Evaluation Expansion

ChatGPT Plus specifically suggested this: once you have enriched company profiles, use your local LLM to generate synthetic Q&A pairs from each profile. This expands your 50 human-validated questions into hundreds of evaluation examples and helps tune retrieval parameters for Georgia-specific queries.

- Prompt: 'Given this company profile, generate 5 realistic industry-user questions and grounded answers. Questions must be answerable from the text only.'
- Store synthetic Q&A in a separate eval dataset — do not mix with production KB
- Use for tuning top_k, chunk size, and prompt variants before final evaluation

10. Expected Impact on Your RAG Evaluation Metrics

RAGAS Metric	Current State	After Phase 1	After Phase 2	After Phase 3	Why
Answer Correctness	Low-Medium	Medium	Medium-High	High	More factual KB

					content to retrieve
Context Recall	Low	Medium	High	High	More relevant documents exist to find
Faithfulness	Medium	Medium-High	High	High	Longer context = less hallucination
Context Precision	Medium	Medium	Medium-High	High	Richer metadata enables better filtering
Answer Relevancy	Medium	Medium-High	High	High	Better descriptions match query intent
Source Attribution	Low	Medium	High	High	Clear OEM/supplier relationships to cite

11. Final Summary — Your Action List

If you only do 5 things, all 6 AI sources agreed these have the highest ROI:

1. Georgia.org press releases + GDEcD announcements → investment amounts, dates, job counts
2. Company websites for each of your 42 EV companies → fix the 46-character description problem
3. Savannah JDA supplier list (savannahjda.com) → solve OEM relationships for Hyundai ecosystem
4. Georgia EPD air permits → facility capacity and process confirmation
5. LLM relationship extraction from press releases → build supplier graph edges

Remember the transformation goal:

Current system answers: **'Who exists?'**

Target system answers: **'Who supplies whom, where, with what, how much, when, and how fast growing?'**

This requires: **Events + Edges + Extended Text + Time + Continuous Updates**